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A comprehensive analysis of spatial utilization in a state transport bus through activity mapping of drivers and passengers.

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This study was undertaken among 100 randomly selected bus conductors from 2 routes. A questionnaire study based on the modified Nordic musculoskeletal questionnaire, assessment of physical and physiological parameters, analysis of working postures and a detailed work study were performed. The analysis revealed that conductors had a work schedule of 16–18 h each day; the duration of work could vary from 15 to 20 days at a stretch. Discomfort leading to musculoskeletal disorders mainly affecting the leg (93.3%), knee (83.3%), shoulder (80%) and back areas (56.7%) had the highest 12-month prevalence rates and increased day by day. The conductors also suffered from extreme physiological stress due to prolonged working hours in hazardous standing posture conditions, excessive work pressure and minimum rest between trips. Consequently, all those factors affected their health and work performance.

1. INTRODUCTION

Public transportation plays a crucial role in modern urban societies, offering a convenient and sustainable mode of travel. Buses, as a primary component of urban transit systems, serve as the lifeblood for countless commuters worldwide. However, the overall health and comfort of passengers and drivers are significantly influenced by the spatial design of the bus, the activities taking place within it, and the associated health risks stemming from ergonomic factors. This research paper aims to investigate the various spaces within a bus, the activities of passengers and drivers, and explore potential interventions for improving bus ergonomics. Moreover, it seeks to illuminate the health risks associated with the spatial utilization and design of buses, thereby paving the way for informed, practical recommendations.

Efficient spatial utilization and ergonomic considerations are not only essential for the comfort of passengers but also for the well-being of bus drivers. Passengers often spend extended periods on buses, making it imperative to analyze how they interact with the available spaces, seating arrangements, and amenities. Additionally, bus drivers face unique ergonomic challenges, as they spend

long hours navigating through urban traffic, which can take a toll on their health. By studying the spatial dynamics within buses and the activities of both passengers and drivers, we can identify areas for improvement and develop evidence-based suggestions to enhance the overall bus experience.

Understanding the intricacies of bus ergonomics and health risks requires a multifaceted approach, encompassing aspects such as seat design, aisle space, safety measures, and driver well-being. This research paper will systematically explore these elements, drawing on a combination of empirical data, expert opinions, and relevant literature. Through a comprehensive analysis of the bus environment and the activities occurring within it, we aim to provide valuable insights into the spatial optimization of buses, thereby contributing to the enhancement of passenger and driver well-being.

In the following sections, we will delve into the specific components of bus ergonomics, the associated health risks, and potential interventions aimed at making bus travel safer, more comfortable, and sustainable. Ultimately, this research paper aspires to foster discussions and innovations that will benefit both the passengers who rely on public transportation and the dedicated drivers who ensure its operation.

2. METHODOLOGY

OBSERVATION

The activities of the Telangana bus drivers and the passengers encompassing actions both inside and outside the bus, were observed for two days. This included ingress and egress, sitting, standing, and movement around the bus, by both drivers and passengers; how the driver accessed the switchboard, side mirror, horn, wiper, ignition, and dashboard controls and so on the analysis aimed to identify potential areas for improvement.

DATA COLLECTION

Measurement

The Telangana bus's interior and exterior were

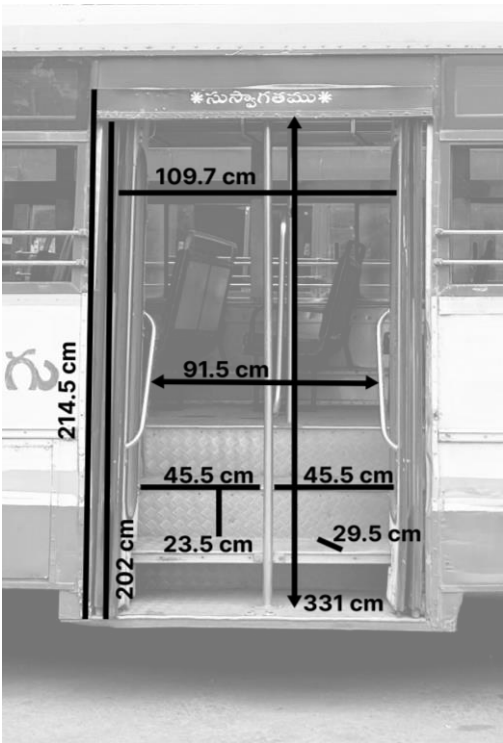
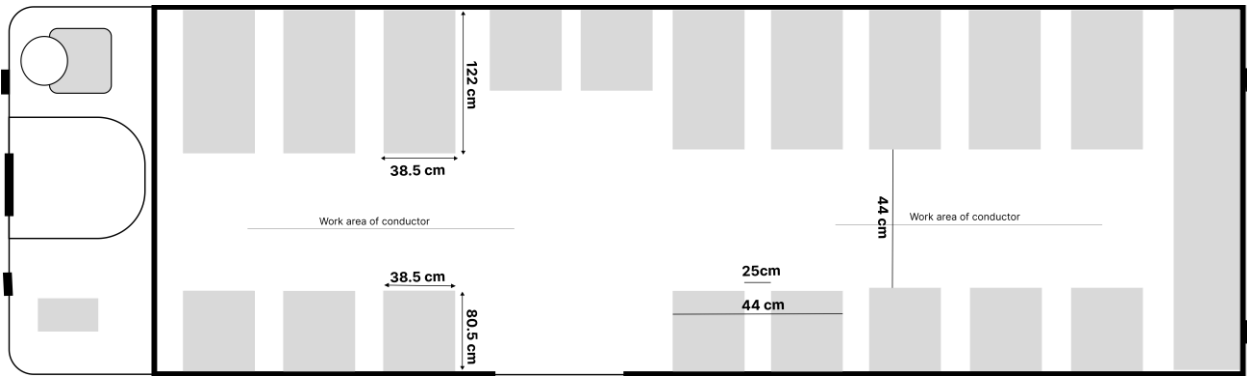
accurately measured using a measuring tape. Subsequently, the anthropometric measures of both drivers and passengers were recorded. The correlation between these two sets of measurements was utilized to pinpoint areas posing potential ergonomic issues.

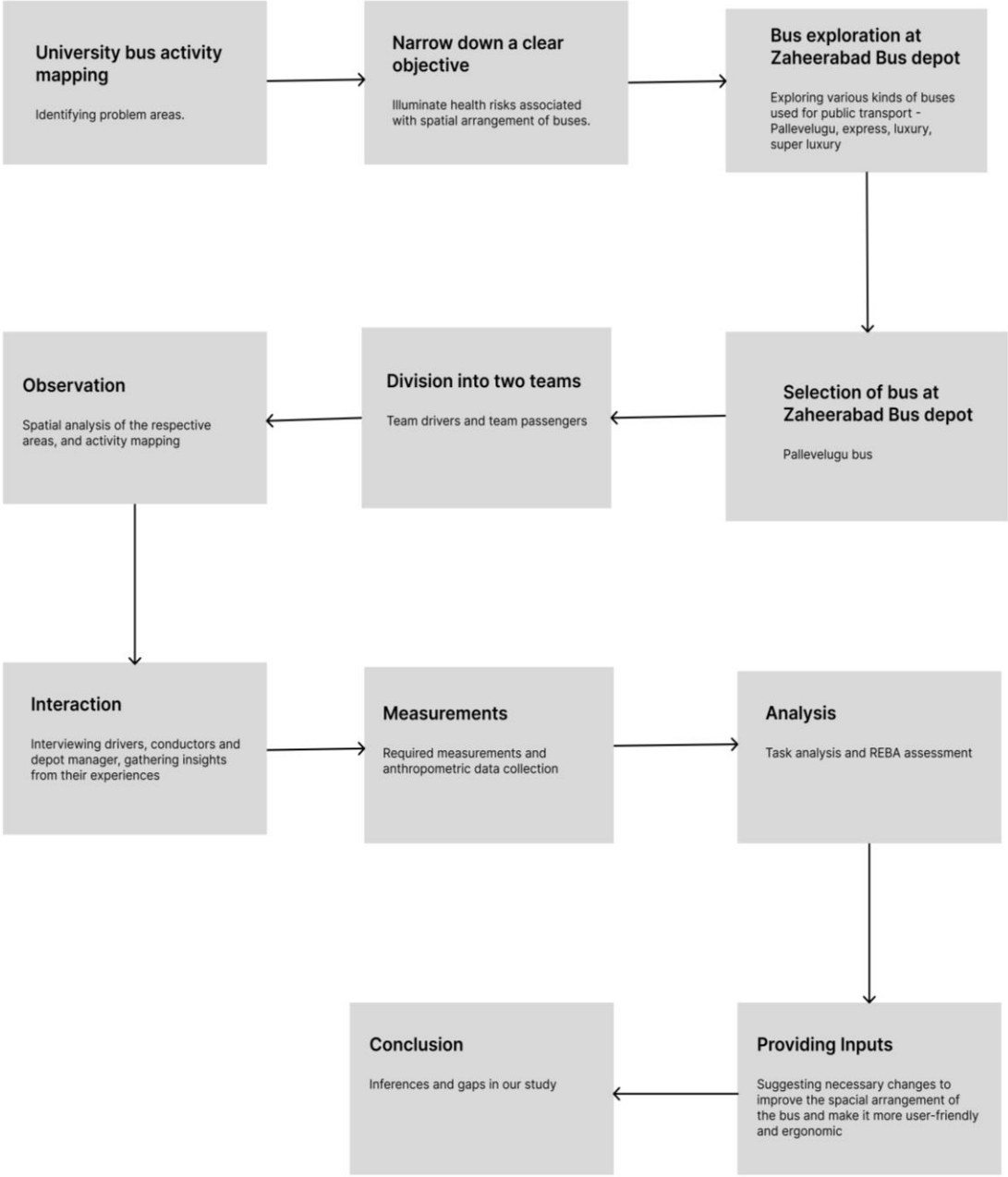
Anthropometric measure checklist (Driver and Passenger)

- Hip height (5)
- Shoulder height (3)
- Stature (1)
- Knee height (15)
- Shoulder - Grip length (25)
- Thumb tip - Grip reach (34)
- Shoulder Breadth-Biacromial (18)
- Sitting eye height (9)

Sitting elbow height (11)
Thigh thickness/clearance (12)
Buttock-Popliteal length (14)
Hip breadth (19)

Elbow-Fingertip length (23)
Body weight (37)
Forward grip reach (36)
Elbow span (33)





TASK ANALYSIS

Driver's space:

From the acquired data of the REBA scores it can be inferred

Using the door to the driver's seat leaves an average REBA score of 11.75 which falls in the high risk category. Driver's door is less in width and height making it difficult to reach out to. Moreover, the driver's themselves rarely use the door meant for the driver's seat as they find it uncomfortable.

The REBA average for reaching the side view mirror is 11.25. The distance between the mirror and driver's seat is the main concern.

The REBA analysis for reaching the steering wheel is 11. Since, the diameter of the steering wheel is of certain length, reaching out to the extreme end of the steering wheel gets difficult.

REBA of seat maximum forward averages to 11.

Seat minimum incline REBA averages to 11.25. The height of the seat and placement of lever is the cause of the body movements.

Passenger's space:

Passengers frequently encounter the challenge of navigating tight spaces as they step up and down when boarding or disembarking from the bus.

This can be particularly difficult for individuals carrying luggage, pregnant women, senior citizens, and those with special mobility needs, often leading to discomfort and safety concerns.

When passengers are in the process of boarding the bus, the average REBA score stands at 11.75, categorizing it as a high-risk activity.

On the other hand, while exiting, the REBA score drops to 7.9, classifying it as a medium-risk activity, indicating the differences in physical strain experienced during these actions.

The limited legroom between seats can result in discomfort during longer journeys and may pose a challenge for taller passengers to find a comfortable seating position.

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(6.29) cm and weight 59 (7.50) kg (Table 1). The mean values of BSA and BMI were 1.66 m² and 20.3 kg/m², respectively, which showed that all conductors had a normal range of BSA and BMI.

TABLE 1. Physical Characteristics of Bus Conductors (N = 100)

Characteristic	Bus Conductors	
	M	SD
Age (years)	34.0	11.57
Height (cm)	163.4	6.29
Weight (kg)	59.0	7.50
BSA (m ²)	1.66	0.15
BMI (kg/m ²)	20.3	2.20

Notes. BSA = body surface area, BMI = body mass index.

Table 2 lists mean blood pressure (before and after a trip) and heart rate (before and after a trip). The analysis of physiological parameters showed that blood pressure increased after a trip, followed by heart rate.

The questionnaire study showed that all workers complained of discomfort in different parts of the body. The subjects reported most discomfort

at rest and at work (Figure 1). In this study, 86.6% workers suffered from severe pain at rest and 66.7% at work.

Figure 2 shows that the predominant type of discomfort was pain throughout the body (100%) followed by tingling (83.3%), numbness (50.0%), stiffness (33.3%) and swelling (10%). According to the respondents, pain was greatest at work and also at rest. Due to prolonged standing, they also experienced tingling and numbness mainly in the leg region, followed by the hand region.

In the questionnaire study, most bus conductors reported experiencing discomfort mainly in the legs (93.3%). They experienced more pain in the left leg than in the right one as their prolonged standing posture in the buses exerted excessive pressure on the left leg. The other regions that were affected included the knees (83.3%), ankles (83.3%), feet (83.3%), shoulders (80%), wrists (70%), hands (63.3%) and lower back (56.7%). Because of carrying heavy side bags (2.0–2.5 kg) on their left shoulders, the discomfort in body parts located on the left side of the body seemed more predominant. Amongst the different body

TABLE 2. Physiological Characteristics of Bus Conductors (N = 100)

Physiological Characteristic		Bus Conductors				t
		Before Trip		After Trip		
		M	SD	M	SD	
Blood pressure (mm/Hg)	systolic	119	5.53	136	6.24	10.19
	diastolic	80	3.50	97	3.91	6.86
Heart rate (beats/min)		82	6.90	96	11.35	4.51

Notes. All parameters show significant difference at $p < .05$.

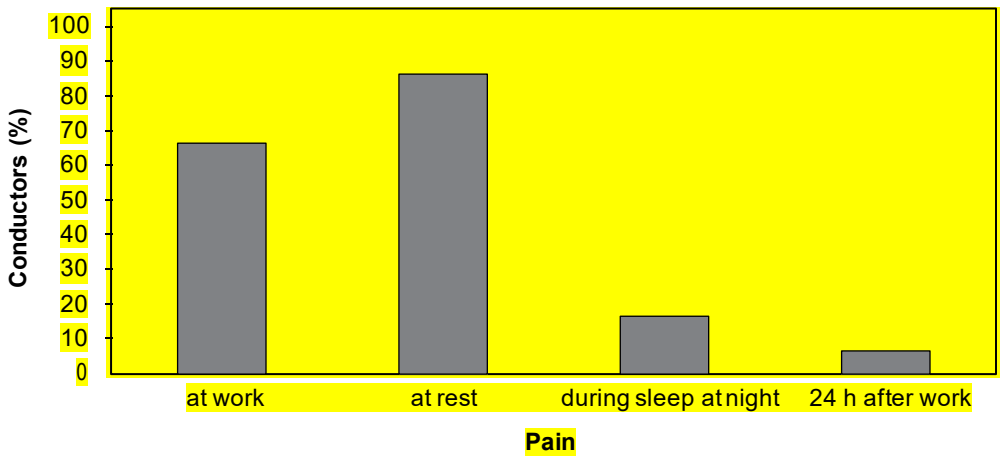


Figure 5. Interior of a Bangla Body bus.

TABLE 6. String Diagram: Study of the Movements of Bus Conductors

TWO-CONDUCTOR PROCESS		SUMMARY							TWO-CONDUCTOR PROCESS	
STUDY OF MOVEMENTS OF CONDUCTOR 1	MOVE-MENT POINTS	METHOD						MOVE-MENT POINTS	STUDY OF MOVEMENTS OF CONDUCTOR 2	
		CONDUCTOR 1			CONDUCTOR 2					
		DIS-TANCE (m)	FRE-QUENCY (times/trip)	TOTAL DISTANCE (m)	DIS-TANCE	FRE-QUENCY	TOTAL DISTANCE (m)			
	A-B	0.9	113	101.7	120	98	117.6	A-B		
	B-C	1.2	142	170.4	30	118	35.4	B-C		
LOCATION LOCAL SHORT-DISTANCE BUS	C-D	0.6	80	48.0	60	102	61.2	C-D	LOCATION LOCAL SHORT-DISTANCE BUS	
	D-E	0.3	180	54.0	120	202	242.4	D-E		
	E-A	1.2	173	207.6	90	142	127.8	E-A		
	TOTAL	4.2	688	581.7	420	662	584.4			

WORK DESCRIPTION CONDUCTOR 1 (FRONT DOOR)	TIME (s)	A	B	C	D	E	E	D	C	B	A	TIME (s)	WORK DESCRIPTION CONDUCTOR 2 (REAR DOOR)
	65.2											91.0	
	42.3											31.2	
	148.2											91.7	
	150.0											170.2	
	10.8											14.4	
	5.5											8.4	
	17.2											22.3	

CONCLUSION

In summary, the study has highlighted pressing ergonomic challenges for both bus drivers and passengers, with uncomfortable seating, reach-related issues, and accessibility problems being major concerns. However, addressing these issues necessitates a careful equilibrium between enhancing comfort and the practicality of modifying bus designs and associated infrastructure. It's important to note that certain factors, such as passenger outfits and age groups, were not considered in the assessment. The absence of female drivers for measurement also presented a limitation. Revamping buses to accommodate a broader range of passenger heights, while minimizing discomfort for the majority, poses a complex task. Altering bus dimensions would have cascading effects on city infrastructure, demanding costly and time-consuming changes. The impact on the urban landscape, public expenditure, and potential disruptions during the redesign process must be thoroughly considered, making this a formidable challenge.